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PC1808

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T.E. (Information Technology)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester- II) (314451)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or 2, Q.3 or 4, Q.5 or 6, Q.7 or 8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume Suitable data if necessary.***

Q1) a) Explain DSDV (Destination Sequenced Distance Vector) with connection establishment and data transfer phase in detail. **[9]**

b) Comment on Adhoc Network MAC Layer with Design Issues, Design Goal. **[9]**

OR

Q2) a) Briefly explain classification of routing protocol for AdHoc Wireless Networks. **[6]**

b) Explain with diagram Clustered Architecture of Sensor Network. **[6]**

c) Write a short note on MACAW. **[6]**

Q3) a) What is stream Cipher? Explain the encryption process using stream Cipher with suitable example. **[8]**

b) What is Cipher Feedback Mode(CFM)? Explain the process of CFM with suitable example. **[9]**

OR

Q4) a) Write comparison between symmetric and asymmetric key cryptography. **[5]**

b) Define Network Attack. Explain with suitable example what do you mean by Active attacks and Passive attacks? **[6]**

c) Describe the following network security threats: **[6]**

i) Unauthorized access

ii) Distributed denial of service

iii) Man in the middle

P.T.O.

Q5) a) Explain PKIX Model. **[9]**

b) Explain Advanced Encryption Standard with diagram. **[9]**

OR

Q6) a) Explain following terms **[9]**

i) Hash Function

ii) Digital Signature

iii) Digital Certificate

b) Explain the working of DES algorithm with suitable example. **[9]**

Q7) a) Define Crime and Cybercrime? State Cybercrimes classification. **[8]**

b) What is the difference between Information security and cyber security? Explain the different layers of security? **[9]**

OR

Q8) a) Explain the concept of Software attacks & hardware attacks with example. **[8]**

b) Write a short note on: **[9]**

i) Malware

ii) Phishing

iii) Cyber Threat

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PC1809

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T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (314452)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***Figures to the right indicate full marks.***

Q1) a) Explain the working of Google File System with a neat diagram with relevant to Big Data. **[9]**

b) Write Short note on YARN and Map - Reduce technique. Is there any difference between both. **[8]**

OR

Q2) a) Explain Hadoop Frame work components with a neat Diagram, which component is used in Big data. **[9]**

b) Write the 4 types of NoSQL databases. How they are used in Big Data. **[8]**

Q3) a) Explain 4 types of big data analytics along with the analytical purpose in each phase. **[8]**

b) Explain different steps in Data Analytics project life cycle, how it is different than normal project life cycle. **[9]**

OR

Q4) a) Explain the tasks in Data integration and Data Transformation phase with suitable example. **[8]**

b) Explain the three ways of Data Normalization used in data analytics. **[9]**

Q5) a) Explain different Data Visualization Techniques. Why visualization is important in Big Data. **[8]**

b) Explain various types of charts used in data visualization along with its analytical purpose. **[10]**

OR

Q6) a) Explain how data Visualization is beneficial for Business Analytics and Intelligence. **[8]**

b) Explain Features of Tableau DV tool and state phases of Tableau workflow. **[10]**

Q7) a) What is Social Media Data Analytics and state the advantages of SMA. **[9]**

b) Explain roles and responsibilities of big data analysts and data scientists. **[9]**

OR

Q8) a) What is Mobile analytics? Explain its importance with suitable example. **[9]**

b) Explain the data analysis life cycle in big data. **[9]**

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PC-1810

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TE-(Information Technology Engineering)

Web App1ication Development

(2019 Pattern) (Semester - II) (314453A)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2 , Q3 or Q4, Q5 or Q6 and Q7 or Q8***
2) ***Neat diagrams must be drawn wtherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume Suitable data if necessary.***

Q1) a) Explain event binding and property binding in angular with example **[6]**

b) What is MVC? Explain MVC architecture in detail **[6]**

c) Explain any 2 Hooks in React JS with example **[6]**

OR

Q2) a) What is TypeScript? Write typescript program which will find maximum and minimum number from array **[6]**

b) Compare functional component with class component **[6]**

c) What is Pipe? Explain with example **[6]**

Q3) a) What is NoSQL? Explain different features of MongoDB **[5]**

b) Write a code using ExpressJS to illustrate the concept of routes **[6]**

c) Explain the concept of Event Loop with suitable diagram **[6]**

OR

Q4) a) What is API? Explain REST HTTP API's in detail **[5]**

b) Write code to create collection, display.and search data in mongoDB using NodeJS **[6]**

c) Explain File system module NodeJS. Write program to perform append and delete operation on file **[6]**

Q5) a) List and Explain any 3 widgets in jQuery Mobile **[6]**

b) What is Mobile-First and Mobile Web? List different Mobile Devices **[6]**

c) What is navigation? Write code to navigate from one page to another in jQuery Mobile **[6]**

OR

Q6) a) Compare jQuery with jQuery Mobile **[6]**

b) Write jQuery Mobile code to hide button once it is clicked **[6]**

c) What is content? Write code to create content in JQuery Mobile page **[6]**

Q7) a) Explain working of Elastic Load Balancer in detail **[6]**

b) What are the different components of VPC? **[5]**

c) What is elastic beanstalk and how it works? **[6]**

OR

Q8) a) What are different AWS Storage types **[6]**

b) What is cloud computing? Enlist benefits of doud computing **[5]**

c) What is elastic beanstalk and how it works? **[6]**

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PC-1811

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T.E. (Information Technology)

ARTIFICIAL INTELLIGENCE (Elective - II)

(2019 Pattern) (Semester - II) (314454 A)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7or Q8*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right side indicate full marks.*
4) *Assume Suitable data if necessary.*

Q1) a) Differentiate between Forward Chaining versus Backward Chaining. **[4]**

b) What is Knowledge Agent? What are the types of Knowledge Agent. Explain any one of it. **[6]**

c) Explain with minimum two examples Predicate Logic and Well-Formed Formula (WFF). **[8]**

OR

Q2) a) Following are some statements, represent it in the form of nodes, arcs and relationship. **[5]**

- Jerry is a cat.
- Jerry is a mammal.
- Jerry is owned by Priya.
- Jerry is brown coloured.
- All Mammals are animal.

b) Differentiate between Propositional versus First Order Predicate Logic.**[6]**

c) Write a note on Bayes Theorem and Bayesian networks. **[7]**

Q3) a) What is NLP? List the steps of in NLP? Explain Lexical Analysis with an example. **[9]**

b) What is Syntactic Parsing and its Constituents with reference to Grammar. Draw the Syntactic Parse tree for the following statements. **[8]**

P.T.O.

i) "The cat chased the mouse."

ii) "John eats an apple."

iii) "The big brown dog barks loudly."

iv) "I saw a black cat on the roof."

OR

Q4) a) Define Ambiguity in reference to Lexical, Syntactic and Semantic phases of NLP? with example. Brief about the Pragmatic Analysis in NLP. **[9]**

b) Describe the general process of building a text classification system for spell checking. Discuss the steps involved, such as data collection, feature extraction, model training, and evaluation with the help of an example. **[8]**

Q5) a) Describe Mini-Max Search Procedure with a suitable example. **[9]**

b) Illustrate Goal-Stack Planning with an example. **[9]**

OR

Q6) a) Explain the concept of alpha-beta pruning in the context of the minimax algorithm. Discuss how alpha-beta pruning improves the efficiency of the minimax algorithm by eliminating unnecessary evaluations of game tree nodes. Provide an example to illustrate the application of alpha-beta pruning in a game tree. Justify, your answers with detailed explanation. **[9]**

b) Describe Reactive Systems with the help of suitable examples. **[9]**

Q7) a) Write a note on (Any Two): **[12]**

i) Convolutional Neural Networks (CNNs)

ii) Deep belief Networks

iii) Restricted Boltzmann Machines

b) Explain in brief AI-Robotics. **[5]**

OR

Q8) a) Write brief note on (Any Two): **[12]**

i) Computer Vision in AI

ii) AI-Neural Networks

iii) AI-IOT

b) Explain in brief Tensor Flow. **[5]**

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PC-1812

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T.E. (Information Technology)
Cyber Security (Elective-Il)
(2019 Pattern) Semester - II (314454B)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Que 1 or Que 2, Que 3 or Que 4, Que 5 or Que 6, Que 7 or Que 8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume Suitable data, if necessary.***

Q1) a) What is cyber forensics? What are the different types of cyber forensics? **[6]**

b) What do you mean by Application Forensics Readiness? Describe any 3-web application forensic tools. **[6]**

c) What is crypto currency? Explain crimes related to bitcoins. **[6]**

OR

Q2) a) What is the process of conducting a cyber-forensic investigation? **[6]**

b) What are the differences between cyber forensics and digital forensics? **[6]**

c) Write any case study on cyber forensics and its Investigation Reports. **[6]**

Q3) a) What are the 5 phases of digital forensics? Explain. **[6]**

b) What are some common data sources for digital forensics investigations? **[6]**

c) What are Advanced Forensic Tools? Explain anyone. **[5]**

P.T.O.

OR

Q4) a) Can you give me some examples of when and where digital forensics might be used? **[6]**

b) Why would someone seek to destroy digital evidence? What techniques can they use? **[6]**

c) Write a detailed case study on Digital Forensics. **[5]**

Q5) a) What are the types of social engineering? **[6]**

b) What are the characteristics of social engineering? **[6]**

c) Explain Phishing Attack with suitable example. **[6]**

OR

Q6) a) What is spear phishing in social engineering? **[6]**

b) What are the phases in social engineering life cycle? **[6]**

c) How to prevent Insider Threats? Explain the process. **[6]**

Q7) a) What is the Cybercrime? How this crime is a distinct from other types of Crime? **[6]**

b) Write the detailed note on the cyber-crime under its IT act? **[6]**

c) Explain the International Standards maintained for Cyber Security and Security Audit. **[5]**

OR

Q8) a) What Is Cyber Stalking? Explain the legal remedies available against The Cyberstalking in India. **[6]**

b) Explain cyber terrorism and provision under the IT Act. **[6]**

c) Explain the positive aspects and weak areas of ITA 2000. **[5]**

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PC-1813

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T.E. (Information Technology)
CLOUD COMPUTING (Elective-Il)
(2019 Pattern) (Semester-II) (314454C)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6,Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) What is the purpose of the Open Cloud Consortium? What are the common standards supported by open cloud Consortium? **[10]**

b) What is the use of Windows Azure? Explain main elements of Windows Azure? **[8]**

OR

Q2) a) Explain the architecture of Google App Engine with neat diagram. **[10]**

b) How are Spot Instance, On-demand Instance, and Reserved Instance different from one another? **[8]**

Q3) a) What are the three important components of HDFS? Enlist the difference between GFS and HDFS. **[9]**

b) Explain the basic components DynamoDB? Explain the advantages of DynamoDB **[9]**

OR

P.T.O.

Q4) a) Describe Google file system with neat diagram. **[9]**

b) Explain various Cloud Computing Security Controls with an example? **[9]**

Q5) a) What are different enabling technologies used in IoT. Explain any one application of Internet of things? **[9]**

b) What is ZigBee Technology. Explain different layers of the ZigBee protocol stack? **[8]**

OR

Q6) a) Explain applications of the Internet of Things used in Retailing and Supply-Chain Management. **[9]**

b) Draw an architecture of RFID along with an application. **[8]**

Q7) a) Describe the use of cloud-based smart fabrics and paints. **[9]**

b) What are the Benefits of Mobile Cloud Computing. How it is different than cloud computing. **[8]**

OR

Q8) a) Explain the Docker architecture with neat diagram. **[9]**

b) Write short note on : i) Mobile Cloud ii) Multimedia Cloud **[8]**